

END SEMESTER EXAMINATION DECEMBER 2025

Name of the Program: B.Tech CSE (AIML)

Semester: III

Name of the Course: Python Programming

Course Code: TCS346

Time: 3 Hours

Maximum Marks: 100

Note:

- All questions are compulsory.
- Answer any two sub questions among a, b & c in each main questions
- Total marks in each main question are twenty.
- Each questions carries 20 marks

Q1	(20Marks)	CO5
(a)	A company named GreenEnergy Pvt. Ltd. wants to visualize its annual energy production from different renewable sources using a pie chart in Python. The data (in Gigawatt-hours) is given below: sources = ['Solar Power', 'Wind Energy', 'Hydropower', 'Biomass', 'Geothermal'] energy_produced = [320, 280, 200, 100, 60] Using Matplotlib, develop a Python program to create a pie chart and customize it using the following parameters: - Display each source as a slice using appropriate labels and colors. - Show percentage values on the slices using autopct='%1.1f%%' and set the pctdistance to 0.6. - Add a shadow to the chart and set the startangle to 90 degrees. - Highlight the highest contributor (Solar Power) using explode=[0.1, 0, 0, 0, 0]. - Set counterclock=False and apply styling with wedgeprops={'linewidth':2} and textprops={'fontsize':12} for better presentation.	
(b)	Describe various array operations supported by NumPy with appropriate code snippets to demonstrate these operations on one-dimensional and two-dimensional arrays.	
(c)	A healthcare organization has collected patient health records from multiple clinics across different regions. The dataset includes details such as patient ID, age, blood pressure, cholesterol level, diagnosis date, and treatment cost. However, during inspection, several data quality issues were found, including missing values, duplicate patient entries, inconsistent date formats, incorrect data types, and a few abnormally high treatment costs. As a data analyst, describe the data cleaning steps you would perform to prepare this dataset for analysis. Explain how you would handle missing values, remove duplicate	

Q2	records, correct data types and date formats, detect and treat outliers. Discuss how these cleaning processes enhance the accuracy, reliability, and usability of the dataset for meaningful healthcare analytics.																
(a)	Explain method overloading and overriding concepts in Python. Develop a Python program to demonstrate method overloading and method overriding using a simple calculator example.	C1															
(b)	Define Inheritance and explain how it helps in code reusability and extension. Describe different types of inheritance supported in Python (single, multilevel, multiple, and hierarchical) with syntax and examples.																
(c)	Discuss the following concepts with code snippets: - def __init__ () constructor - class and objects - super() function in inheritance																
Q3	The academic department of TechVile University wants to visualize the semester-wise SGPA trend of two students to analyze their academic performance over time. The SGPA data for four semesters is given below: <table border="1"> <thead> <tr> <th>Semester</th><th>Dia</th><th>Ayush</th></tr> </thead> <tbody> <tr> <td>1</td><td>7.8</td><td>8.2</td></tr> <tr> <td>2</td><td>8.5</td><td>8</td></tr> <tr> <td>3</td><td>8.9</td><td>8.6</td></tr> <tr> <td>4</td><td>9.1</td><td>8.4</td></tr> </tbody> </table>	Semester	Dia	Ayush	1	7.8	8.2	2	8.5	8	3	8.9	8.6	4	9.1	8.4	C1
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(a)	Using Matplotlib, develop a Python program to plot a line chart comparing both students' SGPA progress over semesters. Apply the following customizations using the given parameters: - Plot both lines using different colors and linestyles ('-', '--'). - Add markers for each data point with appropriate size, face color (mfc), and edge color (mec). - Set the linewidth and alpha (transparency) for each line. - Add labels for each line (label) and display the legend. - Use zorder to ensure one student's line appears above the other for clear visibility.																
(b)	Discuss Pandas merging and grouping operations with code snippets by considering appropriate datasets																
(c)	i. Describe the private, protected, and public members concepts in Python with code snippets.																

ii. Explain the purpose and significance of the self keyword in Python classes. How does self help in distinguishing between instance variables and local variables within a class? Support the explanation with code snippets.

Q4 - Explain what outliers are in a dataset and why it is important to detect and handle them during data analysis. Discuss two commonly used methods for detecting outliers, such as the Interquartile Range (IQR) method and the Z-Score method. Write the formulas for each and explain how they identify values that lie significantly above or below the normal range of data.

A logistics company named FastMove Couriers maintains employee data in a file named delivery_staff.csv. The dataset contains some missing and inconsistent values that require cleaning before analysis.

Emp_ID	Name	Deliveries_Completed	Customer_Rating
1	Ramesh	120	4.8
2	Ananya	95	NaN
3	Suresh	NaN	4.2
4	Meena	150	4.9
5	Kavya	80	3.8
6	Arjun	NaN	NaN

Using Python Pandas, implement a program to perform the following tasks:

- Load and display the dataset.
- Identify and display rows with missing values.
- Replace missing Deliveries_Completed with the mean of existing values.
- Fill missing Customer_Rating with 4.0.
- Create a new column Performance_Level using a lambda function such that:
 - If Deliveries_Completed ≥ 120 and Customer_Rating $\geq 4.5 \rightarrow$ "Excellent"
 - If Deliveries_Completed ≥ 90 and Customer_Rating $\geq 4.0 \rightarrow$ "Good"
 - Otherwise $\rightarrow$ "Needs Improvement"
- Display the cleaned and updated DataFrame.

Discuss the following with code snippets:

- Dictionary data structure
- Exception handling

Q5 - Discuss how positional arguments, keyword arguments, variable-length positional arguments (*args), and variable-length keyword arguments (**kwargs) are used to pass data into functions with suitable examples. Describe the advantages of using *args and **kwargs for creating flexible and reusable functions in Python.

(b) Explain how metacharacters (like ., ^, \$, |, *, +, ?, and {}) and special sequences (such as \d, \w, \s) are used to build complex search patterns. Provide an example for each metacharacter and special sequence to illustrate their use in real-world scenarios.

Develop a Python program to demonstrate various set operations in a real-life scenario. A travel agency named GlobeTrek maintains two sets:

```
summer_packages = {'Goa', 'Kerala', 'Manali', 'Shimla'}
winter_packages = {'Manali', 'Shimla', 'Kashmir', 'Leh'}
```

Perform the following operations using appropriate set methods:

- Display all unique destinations offered by the agency (union).
- Find the destinations common to both summer and winter packages (intersection).
- List the destinations available only in summer but not in winter (difference).
- Check whether winter_packages is a subset of summer_packages.
- Add a new destination 'Ooty' to the summer_packages set and remove 'Leh' from winter_packages.